

ing students and enthusiasts to develop/assemble the board themselves and build innovative and interactive projects around it. An Arduino Duemilanove board is used in this project. It is fully compatible with Open Source Arduino software.

You can also design this project using any of the clones of the Arduino board available in the market. The clone versions are fully compatible with the Arduino IDE and the various Arduino shields, facilitating interfacing with motor, the Internet and much more.

LED matrix. Fig. 3 shows the pin-out of a bi-colour LED. To make the LED glow, cathode (K) terminal should be connected to ground, and anode (A1 or A2) to Vcc. Bi-colour LED is similar to a single-colour LED; the only difference is that it houses two LEDs (one red and the other green) in a single package. Bi-colour LEDs have three leads—a common cathode to which both the LEDs are connected internally, and two anodes (one for each colour).

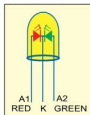


Fig. 3: Pin details of bi-colour LED

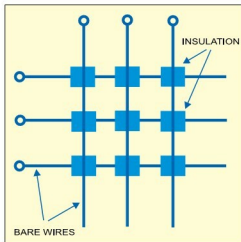


Fig. 4: Touchpad construction

If a dedicated pair of pins was required for each of the eighteen LEDs (nine red and nine green), one would quickly run out of pins on the Arduino board. However, with multiplexing, all the LEDs can be addressed with only nine pins (refer Fig. 2).

To achieve this multiplexing in hardware, the nine bi-colour LEDs are placed and wired in a cross-bar manner in three rows and three columns. Same-colour anodes (nine red and nine green) are grouped together in three columns of three, while the cathodes are grouped in three rows with three common cathodes in each row.

Typically, all the anode and cathode lines are at high impedance. However, an LED can be selected and lit up with the desired colour by setting the corresponding anode line high and cathode line low. For instance, in Fig. 3, if green anode (A2) line is set high and the cathode (K) line is set low, the LED gets illuminated with green colour.

To light up more than one LED at a time, the 'persistence of vision' technique is used. By selecting pairs of LED lines rapidly, an

illusion of all the selected LEDs being lit up together is created. The circuit (refer Fig. 2) shows how the LEDs are physically wired up. The red and green anode lines are connected to the pulse-width modulation pins of the Arduino board (pins 3, 5, 6, 9, 10 and 11) to have effects like fading later on. The cathode lines go into pins 4, 7 and 8. Each of the cathode and anode lines has a 220-ohm current-limiting resistor in series.

Touchpad. The purpose of the touchpad is to activate the LEDs in the 3x3 matrix with the touch of a fingertip. Like the LED matrix, a cross-bar multiplexer arrangement can be used to construct a touch-sensor array or a touchpad. This can be constructed from a sheet of thin acrylic or plastic, provided it is large enough to overlay on the LED matrix.

The touchpad is a simple wire grid with three bare wires running in rows, and three bare wires running in columns above the rows. The row and column wires are taped to the thin acrylic sheet using a transparent tape, which also acts as the insulating spacer between the wires at the intersections (refer Fig. 4). At each intersection point, there is a 4x4mm² insulating tape. A finger-touch at the intersection point makes contact with both the wires. A small but detectable current therefore flows from one wire to the next via the finger. This makes the corresponding LED to light up.

The touchpad is connected to the Arduino board using six analogue pins, with pins 0, 1 and 2 connected to the wire grid rows, and pins 3, 4 and 5 connected to the columns (refer Fig. 2).

Software

The software code is written in Arduino programming language, which is an implementation of Wiring language—a physical computing platform similar to the processing multimedia program-

Arduino Duemilanove Board Features

Board	Features
Microcontroller	ATmega168 or ATmega328
Operating voltage	5V
Input voltage (recommended)	7-12V
Input voltage (limits)	6-20V
Digital input/output (I/O) pins	14 (of which 6 provide pulse-width modulation output)
Analogue input pins	6
DC current per I/O pin	40 mA
DC current for 3.3V pin	50 mA
Flash memory	16 kB (ATmega168) or 32 kB (ATmega328), of which 2 kB is used by boot-loader
SRAM	1 kB (ATmega168) or 2 kB (ATmega328)
EEPROM	512 bytes (ATmega168) or 1 kB (ATmega328)
Clock speed	16 MHz